SAVINDI WIJENAYAKA, Ph.D.

Senior Machine Learning Engineer & Researcher

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Summary

Machine Learning Engineer with a PhD and 6+ years of applied ML experience across industry and research, developing and deploying production-grade deep learning systems in cloud-native environments. Brings technical depth across the entire journey from problem framing and dataset construction through model architecture, training, and system design, to production deployment, owning each phase with a focus on business impact, engineering rigour, and data governance. Driven by the challenge of turning rigorous research into systems that create tangible, real-world impact across diverse domains and data types.

Technical Skills

- **ML & Research:** Computer Vision, NLP, Signal Processing, Biological modelling and computational analysis
- **Languages & Frameworks:** Python, PyTorch, Keras, Flask, Django, Java, Spring Boot
- **Data types:** Unstructured: Text, Image, Video, Audio, 3D, Point Cloud, Mesh; Structured: Time Series, Tabular
- **Data & Cloud Infrastructure:** Azure, AWS, Docker, Kubernetes, Data warehouse (Snowflake), Relational (MSSQL), Time series (Azure Data Explorer), Stream data (Azure Stream Analytics), Transact-SQL
- **Engineering Practices:** Git, GitHub Actions, Azure DevOps Pipelines, REST, gRPC, Agile
- **AI Tooling:** Claude , Cursor

Experience

Intermediate Machine Learning Engineer

Auror · Full-time | Sept 2025 - Present | Auckland, New Zealand

Auror is a retail crime intelligence platform that provides AI-enhanced capabilities to prevent crime.

- Spearheaded scoping, planning, and architecture of an internal RAG-based compliance questionnaire system, driving cross-functional coordination across teams and directing another engineer through full-stack implementation.
- Owned the ML evaluation strategy for flagship entity resolution product, designing a principled multi-gate framework to resolve class imbalance and circular bias, establishing a sound model selection methodology.
- Architected a modular ML evaluation and fairness reporting framework for prospect trials, reducing report turnaround by 80% and accelerating the path to technical win.
- Drove end-to-end modernisation of the entity resolution ML service and architected a region-separated CI/CD pipeline, delivering a new model to production via staged rollout across four regions.
- Established team-wide ML engineering standards via agentic coding conventions and a data licensing review, forming the foundation for a broader data governance framework.

Machine Learning Engineer

WSO2 · Full-time | Sept 2020 - Nov 2021 | Colombo, Sri Lanka

WSO2 is a leading open-source integration vendor; Choreo is its AI-enhanced platform-as-a-service product.

- Researched and delivered a GPT-based model for automated API test case generation, deployed on Azure Kubernetes Service, reducing testing effort for customers and enabling an AI-assisted developer workflow.
- Co-architected a time series-based AI Anomaly Detection system using an event-driven microservices architecture across Azure Stream Analytics, Event Hubs, and Function Apps, enabling customers to detect abnormal application behaviours in real time.
- Implemented a regression testing pipeline for the Ballerina Language Server, systematically surfacing performance degradations and resource utilisation issues to ensure production reliability for users.

- Diagnosed and resolved a critical memory leak in the Ballerina Language Server, applying systematic profiling with JMeter and Eclipse Memory Analyser, to eliminate server crashes and restore resource efficiency.

Software Research Engineer

Pearson · Internship Sept 2018 - Sept 2019 Colombo, Sri Lanka

Pearson is a leading global education provider offering curriculum materials, multimedia learning tools, and testing programs.

- Researched and implemented deep learning modules for emotion detection and speech analysis, integrating models into scalable RESTful microservices for a real-time AI Public Speaking Evaluator.
- Researched and built a Q&A chatbot for Pearson's educational content, iterating from BiDAF to fine-tuned BERT and GPT-2 models to improve answer quality and enable conversational querying of learning material.
- Investigated and developed a flashcard classification service using ULMFiT, LSTM, and GRU to automatically categorise flashcards, reducing manual tagging effort and improving content discoverability for students.

Education

Ph.D. in Bioengineering

University of Auckland Dec 2021 - May 2025 Auckland, New Zealand

Spearheaded an end-to-end deep learning system for automated 3D biological structure analysis, owning every stage from dataset construction and model architecture through ablation studies and the final computational pipeline, resulting in a reproducible workflow applied across 20+ distinct biological samples.

- Engineered a semantic segmentation model with multiscale attention mechanisms, conducting extensive ablation studies to optimise architecture and delivering an automated analysis pipeline that reduced manual processing time by 40+ hours per dataset.
- Developed a Python-based 3D quantification framework grounded in advanced mathematical methods, transforming raw micro-CT data into structured geometric benchmarks and standardising the full workflow from ground-truth labelling to downstream modelling and simulation.

B.Sc. (Hons.) in Software Engineering

University of Kelaniya Feb 2016 - Mar 2020 Kelaniya, Sri Lanka

- Specialised in Data Science and Net-centric application development.
- Attained a GPA of 3.96 out of 4.00, obtaining a First Class.
- Dean's List Honouree, recognised in all four academic years of the B.Sc. programme.

Knowledge Sharing & Technical Outreach

- Member of the teaching team for Code In Place 2021, an online Python course offered by Stanford University, contributing to global tech education initiatives.
- Served as a guest speaker for multiple technical webinars (organised by IEEE and DeepLearning.AI), effectively communicating complex deep learning topics to broader audiences.
- Authored technical articles on Medium covering conceptual topics (CNNs, Kubernetes internals, JVM), practical applications of cloud-native microservices (Kubernetes, Docker, Azure), and automated CI/CD pipelines (GitHub Actions, Azure ARM templates).

Achievements

- **1st Place (2022)** and **2nd Place (2024)** in the international SPARC FAIR Codeathon, representing the University of Auckland, organised by the SPARC Data and Resource Centre and the NIH.
- **4th Place** in DataStorm 2020 Datathon, organised by Octave (JKH) and University of Moratuwa.
- **1st Runner-Up** in National Youth Software Competition 2017, organised by UNDP Sri Lanka.
- **Vice President** of Marketing & Communications at AIESEC in the University of Kelaniya in 2018, contributing to local chapter growth.